

Chapter-14

Fluid Mechanics

- Fluids static's includes hydrostatics, pressure, flotation, Pascal's law and Archimedes principle

$$\text{Relative density [or speciic gravity]} = \frac{\text{Density of substance}}{\text{Density of water at } 4^\circ\text{C}}$$

$$\text{R.D.} = \frac{\rho_s}{\rho_w \text{ (at } 4^\circ\text{C)}}$$

$$\text{Density } (\rho) = \frac{\text{mass}}{\text{volume}} \text{ kg/m}^3$$

$$[\rho] = \text{ML}^{-3}$$

$$\text{Density of mixture} = \frac{\text{Total mass of mixture}}{\text{Total volume}}$$

- (i) If m_1 and m_2 mass of liquid having density ρ_1 and ρ_2 respectively are mixed then

$$\rho_{\text{mix}} = \frac{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}}{\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}} = \frac{\sum m_i}{\sum \left(\frac{m_i}{\rho_i}\right)}$$

$$\text{If } m_1 = m_2, \rho_{\text{mix}} = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} \text{ (H.M.)}$$

- (ii) If v_1 volume of liquid of density ρ_1 and v_2 volume of liquid having density ρ_2 and mixed then

$$\rho_{\text{mix}} = \frac{v_1\rho_1 + v_2\rho_2}{v_1 + v_2} = \frac{\sum(v_i\rho_i)}{\sum v_i}$$

$$\text{If } v_1 = v_2, \rho_{\text{mix}} = \frac{\rho_1 + \rho_2}{2} \text{ (A.M.)}$$

Pressure: Normal force acting per unit area

$$P = \frac{F}{A} = \frac{N}{m^2} \text{ or Pascal (Pa)}$$

- 1 Pa = 10 dyne/cm²
- For a point at depth 'h' below the surface of liquid of density (ρ_l), hydrostatic pressure is

$$p = p_0 + \rho_l gh$$
 where, p = Absolute pressure
 p_0 = Atmospheric pressure
- $(p - p_0)$ is known as Gauge pressure

$$p - p_0 = \rho_l gh$$
 1 atm. = 760 mm of Hg = 1.015×10^5 Pa
 1 torr = 1 mm of Hg
- At a point pressure acts in all direction.
- Pressure always act normal to the fluid boundaries as the ability to flow makes fluid unable to sustain a tangential force.

- The inability of liquid to flow from a vessel having more liquid to vessel having less liquid quantity when both have same level of liquid is called **Hydrostatic paradox**.
- Barometer is used to measure atmospheric pressure while manometer is used to measure pressure difference.

Force of Buoyancy or upthrust

- It is the wt of displaced fluid. It acts vertically upward (opposite to the wt of the body) through centre of gravity of displaced fluid called centre of buoyancy.
- It is independent of all factors of the body such as its mass, size & density etc. except the volume of body inside the fluid.

$$F_B \propto V_{\text{in}}$$
- It depends nature of fluid i.e.,

$$F_B \propto \rho$$
- Its depends on acceleration due to gravity

$$F_B \propto g$$
 so if a lift is accelerated downward ($a_0 < g$)

$$F_B \propto v_{\text{in}} \rho (g - a_0)$$
- Due to upthrust wt of the body will decrease wt of body.
- If w_a is the wt of body in air, ρ_b is the density, v is the volume and w_l is the wt. of body inside liquid then,

$$w_a = v \rho_b g$$
 force of buoyancy $F_B = v \rho_l g$

$$w_l = w_a - F_B$$
 loss in wt. = $w_a - w_l = F_B = v \rho_l g$
- If w_a is weight of body in air and w_w is the wt of body in water then relative density or sp. gravity,

$$\text{sp gravity or R.D.} = \frac{w_a}{w_a - w_w} = \frac{\text{wt in Air}}{\text{Loss in wt in water}}$$
- If w_a is the wt of body in Air, w_w and w_l are wt of the body in water and liquid respectively then sp gravity of liquid,

$$\text{sp. gravity or R.D.} = \frac{\text{loss of wt in liquid}}{\text{loss of wt in water}} = \frac{w_a - w_l}{w_a - w_w}$$
- Volume of cavity in a body (v_c):

If w_a weight in air w_a and w_t in water w_w is given,

$$V_c = \frac{w_a - w_w}{\rho_w} - \frac{w_a}{\rho_b}$$

where weight are in gram.

- If weight w_t of body in air and liquid is given

$$V_c = \frac{w_a - w_t}{\rho_t} - \frac{w_a}{\rho_b}$$

- If an alloy of mass 'm' contains impurity y of density ρ_1 and density of metal is ρ_m . If w_a and w_w are wt of body in air and water then.

$$\frac{(w_a - w_w)}{\rho_w} = \frac{y}{\rho_1} + \frac{(m - y)}{\rho_m}$$

$$\therefore \frac{(m - w_w)}{\rho_w} = \frac{y}{\rho_1} + \frac{(m - y)}{\rho_m}$$

- If wt of body and wt of displaced liquid are equal then body floats in liquid.
- wt. of body = $V_{in} \rho_l g$; $V_{pb} = V_{in} \rho_t$
- If $\rho_t > \rho_b$, then body floats in partially submerged condition.
- If $\rho_t = \rho_b$, then body floats in fully submerged condition.
- App. wt. of floating body is zero.
- If a body floats in a liquid then

- (i) fraction of volume out-side the liquid

$$\frac{V_{out}}{V} = \left(1 - \frac{\rho_b}{\rho_t}\right)$$

$$\% \text{ of volume out side} = \left(1 - \frac{\rho_b}{\rho_t}\right) \times 100$$

- (ii) Fraction of volume inside the liquid.

$$\frac{V_{in}}{V} = \frac{\rho_b}{\rho_t}$$

$$\% \text{ of volume inside} = \frac{100 \rho_b}{\rho_t}$$

- A piece of a ice floats in a liquid of density ρ_t when ice melts completely.
- (i) If $\rho_t > \rho_{water} \Rightarrow$ level rises
- (ii) If $\rho_t = \rho_{water} \Rightarrow$ level remains same
- (iii) If $\rho_t < \rho_{water} \Rightarrow$ level decreases
- If ice containing metal pieces is floating in water and melts completely, level of water falls.
- If ice containing cork pieces is floating in water and melts, level remains same.
- Centre of gravity of floating body and centre of gravity of displaced liquid are on the same straight line known as central line.
- When a floating body is slightly tilted, centre of buoyancy shifts. the vertical line passing through a new centre of buoyancy intersect, the central line at a point called meta centre.
- If metacentre lies above the cg then body in stable equilibrium.
- If metacentre lies below the cg then body is in unstable equilibrium.

- If partially submerged body is floating in liquid is slightly pressed and released, it **oscillates in SHM**.
- If a body is floating in fully submerged is slightly pressed and is released, it **sinks to bottom**.
- If a ball of density ' ρ_b ' is immersed in liquid of density ' ρ_t ' to a depth h and release. The height 'H' above the surface of liquid upto which the ball will jump.

$$H = \left(\frac{\rho_t}{\rho_b} - 1\right) \times h$$

- If the radius of a large bubble becomes n times from the bottom of lake to the surface, the depth of the lake, provided that atmospheric pressure is equal to that of a column of water of height 'H'

$$h = (n^3 - 1) H$$

Principle of continuity:

It is based on conservation of mass.

$$A \times V = \text{constant}$$

- $VR^2 = \text{constant}$

$$A_1 V_1 = A_2 V_2$$

Bernoulli's theorem:

It is based on conservation of energy.

$$P + \rho gh + \frac{1}{2} \rho V^2 = \text{constant}$$

$$\text{Pressure head (z)} = \frac{P}{\rho g}$$

$$\text{Velocity head} = \frac{V^2}{2g}$$

$$\text{Datum head} = h$$

$$\therefore h + \frac{V^2}{2g} + Z = \text{constant}$$

In case of horizontal tube,

$$P + \frac{1}{2} \rho v^2 = \text{constant}$$

Velocity of Efflux:

It is the flow velocity of liquid from an orifice.

$$V_{efflux} = \sqrt{2gh}$$

where h = depth of orifice or hole from surface of liquid.

- Velocity of efflux doesnot depend upon the density of liquid.
- If the level of free surface is at height H from the base and there are two holes at depth 'h' and 'H-h' then range of liquid emerging out are same.
- For maximum range orifice is allowed at half of height of liquid level and maximum range is equal to height of liquid.
- If an orifice of cross-section area A_0 is allowed at the bottom of a tank of cross-section area A then time taken to empty the tank.

$$t = \frac{A}{A_0} \sqrt{\frac{2H}{g}}$$

- Time taken to empty the liquid from H to H' is,

$$t = \frac{A}{A_0} \times \sqrt{\frac{2}{g}} [\sqrt{H} - \sqrt{H'}]$$

Newton's law of viscosity:

$$F = -\eta \cdot A \cdot \frac{dv}{dy}$$

where, F = viscous force

η = coefficient of viscosity

A = Area

$\frac{dv}{dy}$ = velocity gradient

Unit of viscosity (η)

(i) SI unit $\rightarrow$ poiseuli (PI)

(ii) cgs unit $\rightarrow$ poise (P)

$$1 \text{ PI} = 10 \text{ P}$$

- Viscosity depends on the nature of fluid.

(i) In case of liquid viscosity increase with density but for gas decrease with increase in density.

(ii) With rise in temperature the viscosity of liquid decreases while that of gas increases.

(iii) With increase in pressure, viscosity of liquid (except water) increases while that of gas is independent.

- The volume of liquid flowing per unit time is given by

$$\frac{dv}{dt} = \frac{\pi p r^4}{8 \eta L}$$

$$\text{Fluid Resistance, } R = \frac{8 \eta L}{\pi r^4}$$

viscous force (F_v) = $6 \pi \eta r v$

where, r = radius

v = terminal velocity

Terminal velocity,

$$v = \frac{2}{9 \eta} r^2 (\rho - \sigma) g$$

where, ρ = density of body,

σ = density of liquid

r = radius

$V \propto r^2$

$V \propto \rho$

$V \propto \frac{1}{\eta}$

$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2} \right)^2$$

The maximum velocity up to which fluid motion is steady is called critical velocity.

- If $V_{\text{flow}} \leq V_{\text{critical}}$, then flow is stream line or laminar.
- If $V_{\text{flow}} > V_{\text{critical}}$, then flow is known as turbulent.

$$\text{Reynold number } N_R = \frac{V_c \rho r}{\eta}$$

If $N_R \leq 2000 \Rightarrow$ stream line

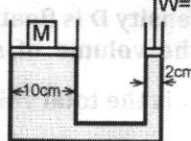
If $2000 \leq N_R \leq 3000 \Rightarrow$ Transition flow

If $N_R \geq 3000 \Rightarrow$ Turbulent flow

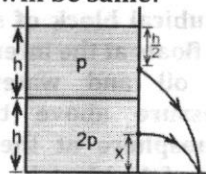
Multiple Choice Questions:

- A ball of mass 1 kg falls from a height of 5 m above the free surface of water. The relative density of the solid ball is $s = \frac{2}{3}$. The ball travels a distance of 2 m under water and becomes stationary. The work done by the resistive forces of water is:
 - 50 J
 - 20 J
 - 40 J
 - 30 J

- A balloon has volume of 1000 m³. It is filled with hydrogen ($\rho = 0.09 \text{ g/litre}$). If the density of air is 1.29 g/litre, it can lift a total weight of:
 - 600 kg
 - 1200 kg
 - 300 kg
 - 1800 kg

- The hydraulic press shown in the figure is used to raise the mass M through a height of 5.0 mm by performing 500 J of the small piston. The diameter of the piston is 10 cm that of the smaller cm. The mass M is
 
 The large one is 2 of:
 - 10^4 kg
 - 10^3 kg
 - 100 kg
 - 10^5 kg

- A container has a small hole at its bottom. Area of cross-section of the hole is A_1 and that of the container is A_2 . Liquid is poured in the container at a constant rate $Q \text{ m}^3/\text{s}$. The maximum level of liquid in the container will be:
 - $\frac{Q^2}{2g A_1 A_2}$
 - $\frac{Q^2}{2g A_1^2}$
 - $\frac{Q}{2g A_1 A_2}$
 - $\frac{Q}{g A_2^2}$

- For what value of x the range on the horizontal ground from both the holes will be same:
 
 - h/2
 - 3h/4
 - h/4
 - they can't be equal

6. Two substances of densities ρ_1 and ρ_2 are mixed in equal volume and the relative density of mixture is 4. When they are mixed in equal masses, the relative density of the mixture is 3. the values of ρ_1 and ρ_2 are:

a. $\rho_1 = 6$ and $\rho_2 = 2$ b. $\rho_1 = 3$ and $\rho_2 = 5$
c. $\rho_1 = 12$ and $\rho_2 = 4$ d. none of these

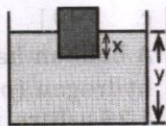
7. A cubical block of wood of specific gravity 0.5 and a chunk of concrete of specific gravity 2.5 are fastened together. The ratio of the mass of wood to the mass of concrete which makes the combination to float with its entire volume submerged under water is:

a. $1/5$ b. $1/3$
c. $3/5$ d. $2/5$

7. A U-tube containing a liquid is accelerated horizontally with a constant acceleration a . If the separation between the two vertical limbs is l , then the difference in the heights of the liquid in the two arms is:

a. zero b. l
c. la/g d. lg/a

8. A cube is floating in liquid. Now temperature is increased considering the expansion of cube only:



a. both x and y will increase
b. x will decrease while y will remain constant
c. both x and y will decrease
d. x will decrease while y will increase

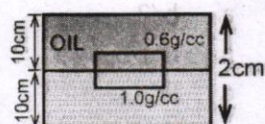
9. A solid of density D is floating in a liquid of density d . If v is the volume of solid submerged in the liquid and V is the total volume of the solid, then $\frac{v}{V}$ equal to:

a. $\frac{d}{V}$ b. $\frac{D}{d}$
c. $\frac{D}{(D+d)}$ d. $\frac{D+d}{D}$

10. A metallic sphere floats in immiscible mixture of water (density 10^3 kg/m^3) and a liquid (density $8 \times 10^3 \text{ kg/m}^3$) such that its $\left(\frac{2}{3}\right)$ part is in water and $\left(\frac{1}{3}\right)$ part in the liquid. the density of the metal is:

a. $\frac{5000}{3} \text{ kg/m}^3$ b. $\frac{10000}{3} \text{ kg/m}^3$
c. 5000 kg/m^3 d. 2000 g/m^3

11. A cubical block of side 10 cm floats at the interface of an oil and water. The pressure above that of atmosphere at the lower face of the block is:



a. 200 N/m^2 b. 680 N/m^2
c. 400 N/m^2 d. 800 N/m^2

12. Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere. Ratio of their volume is:

a. $2:1$ b. $4:1$
c. $8:1$ d. $12:1$

13. Two equal drops of water are falling through air with a steady velocity v . If the drops coalesce, the new velocity will be:

a. $2v$ b. $\sqrt{2}v$
c. $2^{2/3}v$ d. $\frac{v}{\sqrt{2}}$

14. Water flows along a horizontal pipe of which the cross-section is not constant. The pressure is 1 cm of Hg where the velocity is 35 cm/s. At a point where the velocity is 65 cm/s, the pressure will be:

a. 0.89 cm of Hg b. 8.9 cm of Hg
c. 0.5 cm of hg d. 1 cm of Hg

15. The pressure of water in a pipe when tap is closed are $5.5 \times 10^5 \text{ N/m}^2$. When tap gets open, pressure reduces to $5 \times 10^5 \text{ N/m}^2$. The velocity with which water comes out on opening the tap is:

a. 10 m/s b. 5 m/s
c. 20 m/s d. 15 m/s

16. When a large bubble rises from the bottom of a lake to its surface, its radius doubles. The atmospheric pressure is equal to that of a column of a liquid A of height 70 cm. The depth of the lake is (sp. gravity of the liquid = 15.0)

a. 84 m b. 73.5 m
c. 21 m d. 10.5 m

17. A piece of wax weighs xg in air. A piece of metal is found to weight yg in water. It is tied to the wax and both together weight zg in water. Then the specific gravity of wax is ($z < y$):

a. $\frac{x}{y}$ b. $\frac{y}{x}$
c. $\frac{x}{x-(z-y)}$ d. $\frac{x}{x-z}$

18. A large block of ice 5 m thick has a vertical hole drilled through it and is floating in the middle of a lake. The minimum length of the rope required to scoop up a bucket of water through the hole is (density of ice = $0.9 \frac{g}{cc}$):

a. 5.5 m b. 5 m
c. 4.5 m d. 0.5 m

19. Water is flowing through two horizontal pipes of different diameters which are connected together. The diameter of the two pipes are 3 cm and 6 cm respectively. If the speed of water in the narrower

tube is 4 m/s and the pressure is 2.0×10^4 pascal. Then the speed of water in the wider tube is:

- a. 16 m/s b. 1 m/s
c. 4 m/s d. 2 m/s

20. Eight drops of water each of radius 2 mm are falling through air at a terminal velocity of 8 cm/sec if they coalesce to form a single drop. The terminal velocity of the combined drop is:

- a. 32 cm/sec b. 24 cm/sec
c. 16 cm/sec d. 8 cm/sec

21. A drum of radius R full of liquid of density d is rotated about an axis touching its surface at ω rad/sec. The increase in pressure at the centre of the drum will be:

- a. $\frac{\omega^2 R^2 d^2}{2}$ b. $\frac{\omega R d^2}{2}$
c. $\frac{\omega^2 R d}{2}$ d. $\frac{\omega^2 R^2 d}{2}$

22. A rain drop of radius 0.3 mm has a terminal velocity of 1 m/s in air. The viscosity of air is 18×10^{-5} poise. Then the viscous forces on the drop will be:

- a. 16.695×10^{-9} N b. 1.695×10^{-9} N
c. 10.173×10^{-9} N d. 101.73×10^{-9} N

23. A hole is made at the bottom of the tank filled with water (density 1000 kg/m^3). If the total pressure at the bottom of the tank is 3 atmosphere (1 atmosphere = 10^5 N/m^2) then the velocity of efflux is:

- a. $\sqrt{200}$ m/s b. $\sqrt{400}$ m/s
c. $\sqrt{500}$ m/s d. $\sqrt{800}$ m/s

24. Bernoulli's equation is a consequence of conservation of:

- a. energy
b. linear momentum
c. angular momentum
d. mass

25. A tank is filled to a height H. The range of water coming out of a hole which is a depth H/4 from the surface of water level is:

- a. $\frac{2H}{\sqrt{3}}$ b. $\frac{\sqrt{3}}{2} H$
c. H d. $\frac{3H}{4}$

26. There is hole of area A at the bottom of a cylindrical vessel. Water is filled upto a height h and water flows out in 't' second. If water is filled to a height 4 h, it will flow out in time:

- a. $\frac{t}{4}$ b. 2t
c. 4t d. $\frac{t}{2}$

27. An ice-cube of density 900 kg/m^3 is floating in water of density 1000 kg/m^3 . The percentage of volume of ice-cube outside the water is:

- a. 20% b. 35%
c. 10% d. 25%

28. A raft of wood of mass 120 kg floats in water. The weight that can be put on the raft to make it just sink, should be $d_{\text{raft}} = 600 \text{ kg/m}^3$:

- a. 80 kg b. 50 kg
c. 60 kg d. 30 kg

Answers:

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. c | 2. b | 3. a | 4. b | 5. d |
| 6. a | 7. b | 8. c | 9. c | 10. b |
| 11. b | 12. d | 13. c | 14. a | 15. a |
| 16. b | 17. c | 18. d | 19. b | 20. a |
| 21. d | 22. d | 23. b | 24. a | 25. b |
| 26. b | 27. c | 28. a | | |

Solution for Selected Problems

1. $M = 1 \text{ kg}$, $h = 5 \text{ m}$

velocity when it hits the water surface $v = \sqrt{2gh}$

$$v = \sqrt{2 \times 10 \times 5} = 10 \text{ m/sec}$$

Again, Total work done = change in KE

$$w(\text{mg}) + w(\text{F}_B) + w_R = 0 - \frac{1}{2} \times 1 \times 10^2$$

$$\Rightarrow 1 \times 10 \times 2 - \left(\frac{1}{2/3}\right) \times 1 \times 10 \times 2 + w_R = -50$$

$$\therefore w_R = -40 \text{ J}$$

2. Weight lift; $w = v \rho_a g - v \rho_{H_2} g$

$$= 1000 \times 10 (1.29 - 0.09)$$

$$= 12000 \text{ N} = 1200 \text{ kg}$$

3. From conservation of volume,

$$\pi \times \frac{10^2}{4} \times 5 = \pi \times \frac{2^2}{4} \times y$$

$$\therefore y = 125 \text{ mm} = 125 \times 10^{-3} \text{ m}$$

$$w = Fy$$

$$\Rightarrow F = \frac{500}{125 \times 10^{-3}} = 4000 \text{ N} = 400 \text{ Kg}$$

$$\text{Again, } \left(\frac{M_1}{\pi \times \frac{10^2}{4}}\right) = \left(\frac{400}{\pi \times \frac{2^2}{4}}\right)$$

$$\therefore M_1 = 400 \times 25 = 10^4 \text{ kg}$$

4. Let up to height 'h' can be pored

$$(Q - A_1 \times \sqrt{2gh}) = A_2 \times \frac{dh}{dt}$$

$$\text{For } h_{\text{max}}: \frac{dh}{dt} = 0$$

$$\therefore Q - A_1 \times \sqrt{2gh} = 0$$

$$\therefore h = \frac{Q^2}{2gA_1^2}$$

6. When equal volume are mixed then,

$$\rho_{\text{mix}} = \frac{\rho_1 + \rho_2}{2} \Rightarrow \rho_1 + \rho_2 = 8 \dots\dots\dots (i)$$

when equal masses are mixed then,

$$\rho_{\text{mix}} = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} \Rightarrow \rho_1\rho_2 = 12 \dots\dots\dots (ii)$$

Solving (i) & (ii) we get, $\rho_1 = 6$ & $\rho_2 = 2$

$$7. (m + m_c)g = \left(\frac{m}{\rho} + \frac{m_c}{\rho_c}\right) \times \rho_{\text{water}} \times g$$

$$\Rightarrow m + m_c = \frac{m}{0.5} + \frac{m}{2.5}$$

$$\Rightarrow m = m_c \times \frac{1.5}{2.5} \Rightarrow \frac{m}{m_c} = \frac{3}{5}$$

$$10. V\rho_m g = \frac{2}{3}V\rho_w g + \frac{1}{3}V\rho_l g$$

$$\rho_m = \frac{2}{3} \times 1000 + \frac{1}{3} \times 8000$$

$$= \frac{10000}{3} \text{ kg/m}^3$$

11. Pressure of the lower face

$$= \frac{Ah_1\rho_o g + A \times h_2\rho_w g}{A}$$

$$= (0.6 \times 10^3 \times 10 \times 10^{-2} + 1.0 \times 10^3 \times 2 \times 10^{-3}) \times 10$$

$$= (60 + 20) \times 10$$

$$= 800 \text{ N/m}^2$$

12. Excess press inside bubble one; $P_1 = (1.01 - 1) = 0.01$
and $P_2 = (1.02 - 1) = 0.02 \text{ atm}$

$$P \propto \frac{1}{r}$$

$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{P_2}{P_1}\right)^3 = \left(\frac{2}{1}\right)^3 = 8:1$$

$$13. \frac{4}{3}\pi R^3 = 2 \times \frac{4}{3}\pi r^3 \Rightarrow R = 2^{\frac{1}{3}}r$$

$$V_T \propto r^2$$

$$\frac{(V_T)_B}{(V_T)_s} = \left(\frac{R}{r}\right)^2$$

$$\therefore (V_T)_{\text{Bigger}} = 2^{2/3}V$$

$$14. P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$= 1 \text{ cm of Hg} - \left(\frac{1}{2} \times 1000 \times 0.3\right) \times \left(\frac{76}{1.01 \times 10^5}\right) \text{ cm}$$

of Hg

$$= 0.89 \text{ cm of Hg}$$

16. $PV = \text{constant}$

$$P_o \times \frac{4}{3}\pi(2r)^3 = (P_o + H\rho_w g) \times \frac{4}{3}\pi r^3$$

$$\Rightarrow 7P_o = \pi\rho_w g$$

$$\Rightarrow 7 \times h \rho g = H \rho_w g$$

$$H = 7 \times 0.7 \times 15$$

$$H = 73.5 \text{ m}$$

18. Wt of ice block = upthrust

$$\Rightarrow V\rho_{\text{ice}} g = V_{\text{in}} \times \rho_w g$$

$$\Rightarrow (4 \times 5) \times 0.9 \times g = A \times \ell \times 1 \times g$$

$$\ell = 5 \times 0.9 = 4.5 \text{ m}$$

Ice block of length 4.5m is inside the water. Hence minimum length of rope

= length outside the water

$$= (5 - 4.5) \text{ m} = 0.5 \text{ m}$$

$$19. r_1 = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}$$

$$r_2 = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$$

From continuity equation,

$$A_1 V_1 = A_2 V_2$$

$$V_2 = \left(\frac{r_1}{r_2}\right)^2 \times v_1 = \frac{1}{4} \times 4 = 1 \text{ m/sec}$$

$$20. (V_T)_{\text{Bigger}} = n^{2/3} \times (V_T)_{\text{smaller}} \\ = 8^{2/3} \times 8 \text{ cm/sec} = 32 \text{ m/sec}$$

$$21. \frac{1}{2}\rho v^2 = P$$

$$P = \frac{1}{2}d \times R^2\omega^2 \quad \therefore P = \frac{\omega^2 R^2 d}{2}$$

$$22. \eta = 18 \times 10^{-5} \text{ poise} = 18 \times 10^{-6} \text{ poise??}$$

$$r = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$$

$$V_T = 1 \text{ m/sec}$$

$$\text{Viscous force: } F_r = 6\pi r \eta V_T$$

$$= 6 \times 3.14 \times 0.3 \times 10^{-3} \times 18 \times 10^{-6} \times 1$$

$$= 101.73 \times 10^{-19} \text{ N}$$

$$23. \frac{1}{2}\rho v^2 + P_o = 3P_o$$

$$\Rightarrow \frac{1}{2}\rho v^2 = 2P_o$$

$$\Rightarrow \frac{1}{2} \times 1000 \times v^2 = 2 \times 10^5 \quad v = \sqrt{400} \text{ m/sec}$$

$$25. V = \sqrt{2g \times \frac{H}{4}} = \sqrt{\frac{gH}{2}}$$

$$\text{Time of flight } (t_f) = \sqrt{2g \times \frac{(3H/4)}{g}} = \sqrt{\frac{3H}{2g}}$$

$$R = v \times t_f = \sqrt{\frac{gH}{2}} \times \sqrt{\frac{3H}{2g}} = \sqrt{\frac{3}{2}}H$$

$$26. t = \frac{A}{A_o} \times \sqrt{\frac{2H}{g}}$$

$$t \propto \sqrt{H}$$

$$\frac{t_2}{t_1} = \sqrt{\frac{4h}{h}}$$

$$\therefore t_2 = 2t$$

$$27. v_{\text{Pice}} g = V_{\text{in}} \times \rho_w g$$

$$\Rightarrow \frac{V_{\text{in}}}{v} = \frac{\rho_{\text{ice}}}{\rho_w} = 0.9$$

$$\frac{V_{\text{out}}}{v} = (1 - 0.9) = 0.1$$

$$\Rightarrow \% \frac{V_{\text{out}}}{v} = 0.1 \times 100 = 10\%$$

$$28. (m_{\text{wood}} + m)g = v\rho_w g$$

$$\Rightarrow 120 + m = \left(\frac{120}{600}\right) \times 1000$$

$$m = (200 - 120) \text{ kg} \quad m = 80 \text{ kg}$$

